

programs. McCarthy decided that this was just the tip of the solution – the deeper necessity lay in finding out the essence of reasoning and logic.

A lecturer in MIT tried to explain it with the use of a riddle. Let's try it out. In a large maze, there are two different paths to reach the finish line on the other side. Each path poses a different hurdle – one is longer and safer, the other is shorter and more treacherous. Which of the two paths would you choose and why? Got an answer in your head?

The students in the class were asked one by one what they chose and their reason for doing so. Invariably, there were several different thought patterns emerging. And that's when the professor sums up the lesson: No matter which option you go for, you'll end up reaching the same finish line. It doesn't matter when you reach it or how you reached it, since that wasn't the part of the problem.

Similarly, logic works in a way where each problem can have several different ways of solving it. McCarthy wanted to study the essence of abstract reasoning and problem-solving to emulate this ability. For there to exist artificial intelligence, he contended, the machine needed the ability to think for itself while basing the foundation of the thought process on the core of the human reasoning process. To put it differently, he didn't want to emulate a certain “human algorithm of reasoning” in digital format, he wanted to emulate the ability of human reasoning itself.

To this end, McCarthy dove into the world of logic and came up with several pioneering theories and applications, in the process solving some monumental problems in the world of knowledge representation. Given his ability to break down the core concepts of human psychology, it's little wonder then that this is the man who came up with the term “artificial intelligence” itself.

Knowledge and intelligence

McCarthy's reliance on logic and Simon's reliance on the technique to solve problems both had one common hindrance: knowledge. At the time they were conducting their experiments, the storage capacities of computers severely limited their options. Indeed, their entire line of thinking could be said to be a workaround for the limitations of storage space.

But in the human brain, storage space is one of the most important aspects of how we function. Knowledge is power, goes the old adage, after all. The human brain stores massive amounts of information in a physically small space, thus making it an exponentially useful tool. This